

Curriculum Vitae

Shubham Gupta

AI/ML Engineer | Data Scientist
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Skills and Expertise

- **Programming and Databases:** Python, SQL, Neo4j, PostgreSQL, MySQL, Vector Stores (FAISS, Pinecone, ChromaDB)
- **Data Analysis and Visualization:** Pandas, PySpark, NumPy, Power BI, Statistics, Probability, Data Visualization, Time Series Forecasting, ETL, Prefect, Apache Airflow, Git/GitHub
- **Machine Learning and MLOps:** Pytorch, Scikit-learn, TensorFlow, Keras, MLflow, Tensorboard, DVC, CI/CD, Docker, Wandb
- **AI and Deep Learning:** NLP, Transformers, Transfer Learning, RAG, LangChain, langgraph, OpenCV, Neural Networks, Predictive Modeling, Image Processing, Data Augmentation, Feature Engineering, Annotation, LLMs (OpenAI, OpenRouter, Gemini), n8n
- **Cloud & Deployment:** AWS, FastAPI, Streamlit, REST APIs
- **Professional Skills:** Problem Solving, Communication, Collaboration, Adaptability, Time Management, Technical Writing

Projects

FMCG Demand Forecasting & Expiry Reduction

- Built an end-to-end demand forecasting pipeline for a mid-sized FMCG exporter using **Prophet** and **SARIMAX**, with per-SKU seasonal decomposition.
- Delivered a multi-page BI dashboard surfacing SKU-level demand patterns, cutting product expiry losses by **34% within 6 months**.
- Enabled proactive procurement 4–6 weeks ahead, restoring fill-rate compliance with two lapsed distribution partners.

Property Valuation Intelligence System (Australia)

- Trained a **CatBoost + Isotonic regression** AVM on spatial, macroeconomic, and migration signals, achieving **8–12% MAE** vs. the 15% industry baseline.
- Layered a **confidence-interval output** (low / estimate / high) giving investors a negotiation range — up to **\$140K error reduction** on \$2M decisions.
- Implemented continuous retraining on live employment, rental-yield, and migration data to keep valuations market-current.

Production Voice AI Agent — Regional Electricity Utility

- Deployed a fully **on-premise voice agent** for a 50K-household utility; trained a custom regional-dialect TTS model from scraped audio with zero cloud dependency.
- Integrated **real-time tool-access pipelines** for balance lookup, complaint logging, and ID verification against local databases with sub-2s response latency.
- Handled **40% of inbound contact volume** autonomously within month one; **70%** of test users did not identify the system as automated.

Synthetic Supply Chain Data Generator

- Engineered a privacy-safe synthetic data generator using **KDE** for marginal distributions and **OLS regression** to preserve inter-feature correlations from a real supply chain dataset.
- Validated statistical fidelity via **Kolmogorov-Smirnov tests** across all key features; synthetic distributions passed at $p > 0.05$ against real data.
- Enabled downstream ML model training and BI dashboard testing without exposing confidential client inventory records.

Experience

- **Freelancer**
Data Scientist & AI Engineer Remote
Nov 2023 – Present
 - **Client Delivery:** Delivered end-to-end ML solutions across FMCG, PropTech, and Energy sectors for clients in India, Australia, and the UK — achieving a **34% reduction in expiry loss**, **40% autonomous inbound call handling**, and property valuations at **8–12% MAE** vs. 15% market baseline.
 - **Forecasting & BI:** Engineered demand forecasting pipelines (Prophet, SARIMAX) and multi-page Power BI dashboards with SKU-level seasonality analysis, enabling procurement decisions **4–6 weeks earlier** than before.
 - **AI Agents & Voice:** Built production-grade on-premise voice agents with custom regional TTS (trained from scratch), real-time database tool-access, and sub-2s latency — fully air-gapped per client data policy.
 - **MLOps & Deployment:** Maintained ML lifecycle using **MLflow**, **Docker**, and **FastAPI**; implemented continuous model retraining triggered by data drift across forecasting and valuation systems.
- **EvoAstra Ventures | Smarg Technologies**
Data Science Intern Hybrid
Apr 2023 – Sep 2023
 - **NLP & Deep Learning:** Fine-tuned **TinyBERT** on de-identified clinical narratives for disease classification, achieving **87% accuracy** via domain adaptation and optimized text preprocessing.
 - **Analytics:** Built Power BI dashboards and performed structured EDA on **100K+ records**, surfacing actionable patterns for business and operations teams.

Coursework and Achievements

- **FreeCodeCamp** Remote
Machine Learning with TensorFlow 2023
- **365Data Science** Udemy
Data Science with Python 2022-2023
- **DAVV University** Indore, India
Bachelor of Business Administration July 2019 - June 2022
- **Indore Institute of Management and Research** Rau, India
Data Analysis with Python July 2021 - Oct 2021